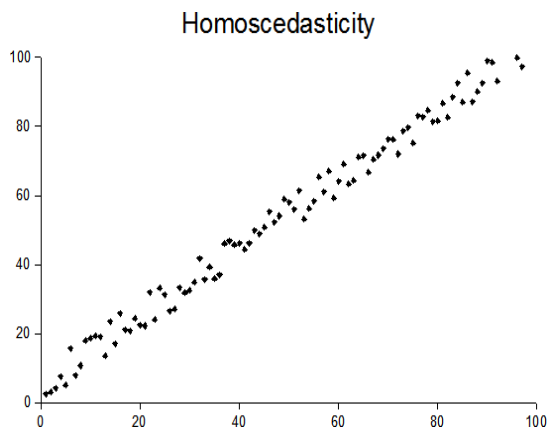
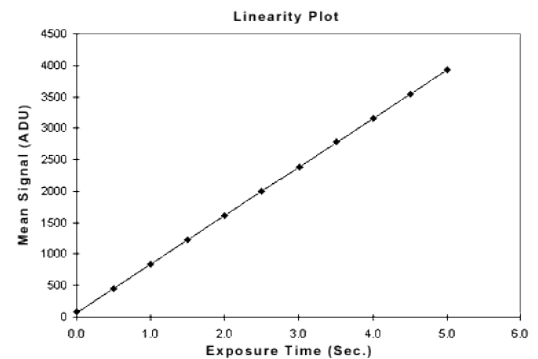


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ASSUMPTIONS OF LINEAR REGRESSION:

Linearity:

- Linearity is the property of a mathematical relationship that can be graphically represented as a straight line.
- Linearity is closely related to proportionality.
- Examples: linear relationship of voltage and current (Ohm's law), and the relationship of mass and weight.

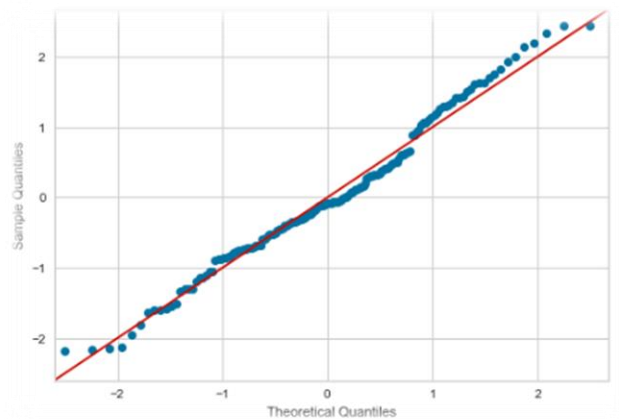


Homoscedasticity:

- The property of having equal statistical variances.
- The error term is same across different values of independent features, regardless of the values of the predictor variables.
- This is also known as homogeneity of variance.

Multivariate Normality:

- The error terms are normally distributed, i.e. the mean of error terms is zero and the sum of error terms is also equal to zero.
- As the sample size goes high, the normality assumption for the residuals is not needed anymore.

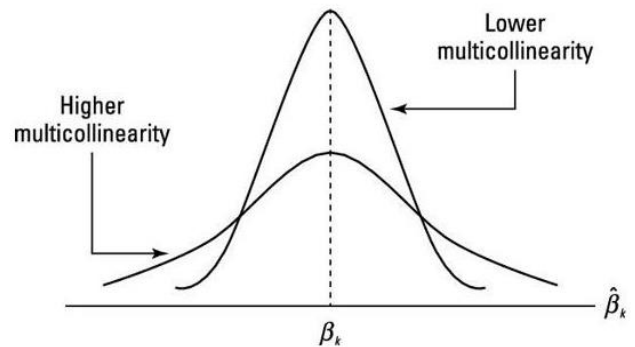


Lack of MultiCollinearity:

- **Multicollinearity** is a state of very high intercorrelations or inter-associations among the independent variables. It is therefore a type of disturbance in the data, and if present in the data the statistical inferences made about the data may not be reliable.

- **Multicollinearity may be tested with three central criteria:**

- Correlation matrix
- Tolerance
- Variance Inflation Factor (VIF)
- Condition Index



- **There are certain reasons why multicollinearity occurs:**

- It is caused by an inaccurate use of dummy variables.
- By the inclusion of a variable which is computed from other variables in the data set.
- Multicollinearity can also result from the repetition of the same kind of variable.
- Generally occurs when the variables are highly correlated to each other.

Independence of Error:

- The assumption of independent errors states that errors should not be correlated for two observations.
- The errors are the deviations of an observed value from the true function value

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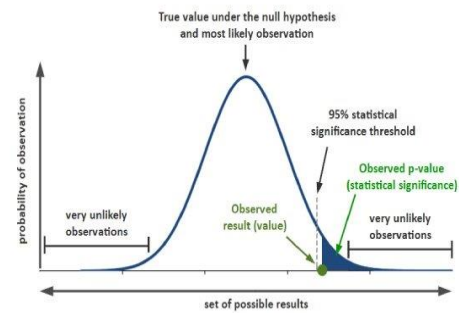
- Null hypothesis: It is regarding the assumption that there is no anomaly pattern or believing according to the assumption made.
- Alternate hypothesis: Contrary to the null hypothesis, it shows that observation is the result of real effect.

P VALUE:

- It is used to determine if the data is statistically incompatible with the null hypothesis i.e P-value gives us the probability of finding an observation under an assumption that a particular hypothesis is true.

$$P \text{ Value} = P(E | H_0)$$

- A sufficiently low value is required to reject the null hypothesis.
- This probability is used to accept or reject that hypothesis.



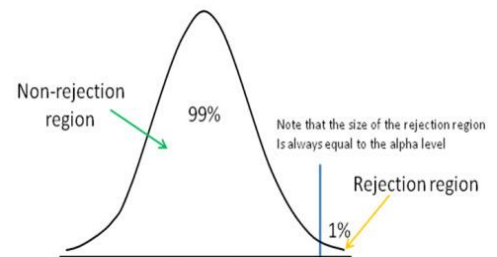
SIGNIFICANCE LEVEL:

- The significance level is the probability of rejecting the null hypothesis when it is true.
- The significance level, also denoted as alpha or α , is a measure of the strength of the evidence that must be present in your sample.
- If the p-value is less than your significance level, you can reject the null hypothesis and conclude that the effect is statistically significant.

$$\text{Significance Level} = p (\text{type I error}) = \alpha$$

- To Find Level of Significance:

- If $p > 0.1$, then there will be no assumption for the null hypothesis
- If $p > 0.05$ and $p \leq 0.1$, it means that there will be a low assumption for the null hypothesis.
- If $p > 0.01$ and $p \leq 0.05$, then there must be a strong assumption about the null hypothesis.
- If $p \leq 0.01$, then a very strong assumption about the null hypothesis is indicated.



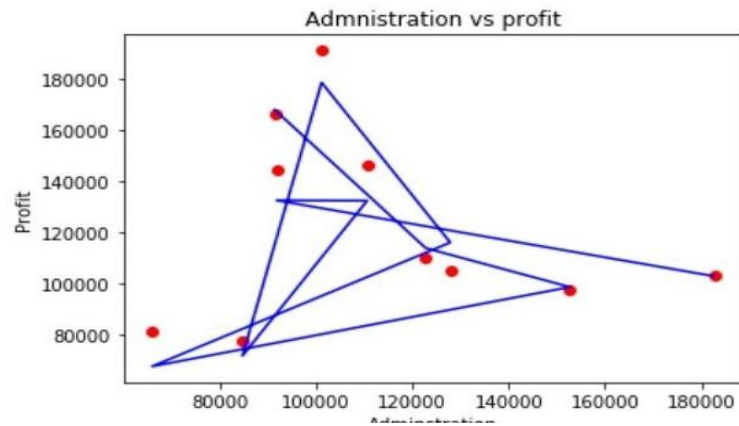
Finding r2_score, Mean Squared Error , Mean Absolute Error:

```
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
r2=r2_score(y_test,y_pred)
mae=mean_absolute_error(y_test,y_pred)
mse=mean_squared_error(y_test,y_pred)
rmse=np.sqrt(mse)
print("r2 = ",r2)
print("mae = ",mae)
print("mse = ",mse)
print("rmse = ",rmse)
```

```
r2 = 0.946458760778722
mae = 6772.453280477903
mse = 68473440.7190593
rmse = 8274.868018225989
```

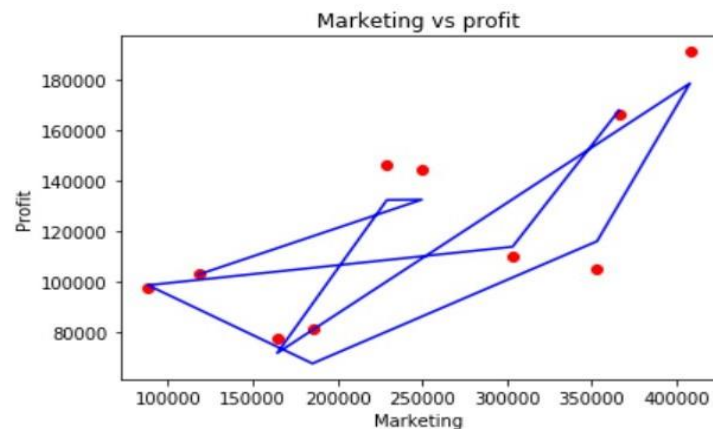
Administration Vs Profit:

```
▶ plt.scatter(x_te[:,4],y_te,color='red')
plt.plot(x_te[:,4],y_pr,color='blue')
plt.title("Administration vs profit")
plt.xlabel("Administration")
plt.ylabel("Profit")
plt.show()
```



Marketing Spend Vs Profit:

```
▶ plt.scatter(x_te[:,5],y_te,color='red')
plt.plot(x_te[:,5],y_pr,color='blue')
plt.title("Marketing vs profit")
plt.xlabel("Marketing")
plt.ylabel("Profit")
plt.show()
```



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AUTOMATIC BACKWARD ELEMINATION :

```
x_1 = x[:,1:]
```

```
const = np.ones((50,1)).astype(int)
```

```
print(const)
```

```
x_1 = np.append(arr = const, values= x_1,axis = 1)
```

##Automatic backward elimination using for loops

```
import statsmodels.api as sm
x_opt=np.array(x_1[:,[0,1,2,3,4,5]],dtype=float)
size=len(x_opt[0])
for i in range(0,size):
    model=sm.OLS(endog=y,exog=x_opt).fit()
    max_p=max(model.pvalues).astype(float)
    if max_p>0.05:
        for j in range(0,size-i):
            if(model.pvalues[j].astype(float)==max_p):
                x_opt=np.delete(x_opt,j,axis=1)
model.summary()
```

1]:

OLS Regression Results

Dep. Variable:	y	R-squared:	0.947			
Model:	OLS	Adj. R-squared:	0.945			
Method:	Least Squares	F-statistic:	849.8			
Date:	Sun, 02 Aug 2020	Prob (F-statistic):	3.50e-32			
Time:	08:11:21	Log-Likelihood:	-527.44			
No. Observations:	50	AIC:	1059.			
Df Residuals:	48	BIC:	1063.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	4.903e+04	2537.897	19.320	0.000	4.39e+04	5.41e+04
x1	0.8543	0.029	29.151	0.000	0.795	0.913
Omnibus:	13.727	Durbin-Watson:	1.116			
Prob(Omnibus):	0.001	Jarque-Bera (JB):	18.536			
Skew:	-0.911	Prob(JB):	9.44e-05			
Kurtosis:	5.361	Cond. No.	1.65e+05			